

Metric bolt torque chart: 8.8, 10.9, 12.9 (M3 to M36)

1. Chart: torques and preloads, M3 to M36 ($\mu = 0.12$)

ISO metric coarse thread, friction of 0.12 under the head and in the thread (oiled steel, the most common case). Torques in newton-metres, preloads in kilonewtons.

Bolt	Pitch	8.8 N·m	10.9 N·m	12.9 N·m	8.8 kN	10.9 kN	12.9 kN
M3	0.5	1.3	1.9	2.2	2.5	3.7	4.4
M4	0.7	3	4.4	5.1	4.4	6.5	7.6
M5	0.8	5.8	8.6	10	7.2	11	12
M6	1	10	15	17	10	15	18
M8	1.25	25	36	42	19	27	32
M10	1.5	48	71	83	30	43	51
M12	1.75	84	123	144	43	63	74
M14	2	133	196	229	59	87	102
M16	2	206	302	354	81	119	139
M20	2.5	416	592	693	130	186	217
M24	3	714	1018	1191	188	268	313
M27	3	1051	1496	1751	246	351	411
M30	3.5	1429	2035	2382	300	427	500
M36	4	2483	3537	4139	438	624	730

They match fastener manufacturers' published charts to within rounding.

Reference values taken from the article <https://www.baouch.fr/en/blog/couple-serrage-vis-tableau/>, which explains the assumptions, formulas and edge cases. For critical use, check the current standard or the manufacturer's documentation. Free to print and share with credit to the source.